

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch : Civil Engineering

Semester :V

Subject Code & Name: BTCVC 501 Design of Steel Structures

Max Marks: 60

Date:28/01/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use of **IS800, IS 875** and **steel table** is permitted

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|---|-------------|----------|
| A) Explain Different types of Loads acting on steel Structure | Knowledge | 6 |
| B) Determine the design strength of rivet in a butt joint by 2 plates 12 mm thick by using 8 mm thick cover plate with hand driven rivets for the case of single cover butt joint and double cover butt joint | Application | 6 |
| C) Explain Different types of riveted joint | Remember | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|---|-------------|----------|
| A) A single angle ISA 90 x 60 x 8 mm is connected with longer leg to the gusset plate of 10 mm thick. Find the effective area
a)with 18 mm diameter rivet
b) welded connection | Application | 6 |
| B) 2 ISA 75 x 75 x 10 mm connected to gusset plate 12 mm thick with 16 mm diameter bolt find the permissible strength in axial tension connected same side of gusset plate take $f_y = 250$ Mpa | Application | 6 |
| C) Calculate moment resisting capacity of a simply supported beam consist of ISMB 300 over a span of 3 m also calculate safe udl(excluding self weight) the beam can carry | Application | 6 |

Q. 3 Solve Any Two of the following.

- | | | |
|---|----------|----------|
| A) Calculate the Maximum wheel load and moments on Gantry girder for the following data
crane capacity = 100 KN
self wt. of crane 100 KN
Self wt of trolley , motor and hook 20 KN
Appx. min approach of crane = 12 m | Analysis | 6 |
|---|----------|----------|

wheel base = 3.0 m

c/c distance between gantry rails = 14 m

span of GG 6 m

self wt of rail section 300 N/m

f_y 250 N/mm²

- | | | |
|--|----------|----------|
| B) Calculate the section modulus and selection of section on Gantry girder for the same data as available on Q 3 A | Analysis | 6 |
| C) Determine the live load per panel point for a Pratt truss of span 15 m with sloping angle 22° take a weight of AC sheet roof covering = 175 N/m ² | Remember | 6 |

Q.4 Solve Any Two of the following.

- | | | |
|---|-------------|----------|
| A) Explain the Lacing system and battening system for columns | Knowledge | 6 |
| B) Write short Note on Grillage Foundation | Synthesis | 6 |
| C) Design a slab base for column section ISHB 300 @63.0 kg/m subjected to axial load of 900 KN M 20 concrete is used for foundation. Provide welded connection between column and base plate | Application | 6 |

Q. 5 Solve Any Two of the following.

- | | | |
|---|-----------|----------|
| A) Explain idealized stress strain curve for mild steel | Remember | 6 |
| B) Explain the concept of plastic hinge | Knowledge | 6 |
| C) Draw with neat sketch of different collapse mechanism | Synthesis | 6 |

***** End *****